

EDWARD PRASCHIFKA

Graduate Software Engineer

Adelaide | 0420 325 252 | edwardpraschifka@gmail.com | github.com/edwardpraschifka | <https://edwardpraschifka.github.io/>

Graduate software engineer with a Bachelor of Mathematical and Computer Sciences, bringing ground-up technical depth together with real-time systems experience from an embedded software engineering internship at a defence technology company. Recently returned from a year of study in China, seeking an entry-level software engineering role.

SKILLS

Programming Languages	Python, C, C++, JavaScript
Frameworks & Databases	React, Express, Node.js, MongoDB, Flask, Django
ML & AI	Machine learning fundamentals, NumPy, Pandas
Math and Statistics	Linear algebra, multivariable calculus, probability, statistics

PROJECTS

TinyVector *August 2026 – September 2026*

(<https://github.com/edwardpraschifka/tiny-vector>)

- Built a from-scratch, templated C++ vector class implementing the full Rule of Five (copy/move constructors and assignment, destructor) with explicit self-assignment protection.
- Enforced const-correctness across all accessors and provided both bounds-checked and unchecked element access, mirroring STL performance/safety conventions.
- Developed a Catch2 test suite covering construction, copy/move semantics, and arithmetic operators, validated under AddressSanitizer to catch memory errors at the point of occurrence.

Orbit *May 2026 – August 2026*

(<https://github.com/edwardpraschifka/orbit>)

(<https://orbit-upv6.onrender.com/>)

- Built a full-stack pipeline that pulls raw data from the Steam Web API, transforms it, and serves it as an interactive 3D force-directed graph of a user's friend network.
- Designed a modular Flask backend (routes, services, graph, cache, metrics) with in-memory caching and rate limiting to manage API load and keep data flow efficient.
- Built incremental graph expansion, turning raw external API data into a clear, structured, navigable dataset.

EXPERIENCE

Embedded Software Engineer Intern — Silentium Defence, Adelaide *Aug 2024 – Dec 2024*

- Conducted a cost-benefit analysis comparing two candidate solutions for sending real-time location data from a drone, weighing each against task requirements to inform the team's decision.
- Authored a comprehensive setup report enabling easy replication of the solution by other team members.
- Configured a Raspberry Pi 5 with the required OS, software, and hardware to reliably forward received location data to endpoints.

EDUCATION

Bachelor of Mathematical & Computer Sciences — The University of Adelaide *Jan 2020 – Dec 2024*

Diploma of Languages (Mandarin) — The University of Adelaide *Jan 2020 – Dec 2024*

LANGUAGES & AWARDS

- Mandarin — HSK 5 (~2,500+ vocabulary), government-standard proficiency exam; used effectively in professional and academic contexts.
- CIW China Studies Fellowship (Feb 2025 – Feb 2026) — fully immersive year-long study program in China for Australian graduates.

VOLUNTEERING

- Kitchen Assistant, Meals on Wheels (Aug 2022 – Aug 2024) — prepared and delivered meals to hundreds of elderly clients as part of a small team.